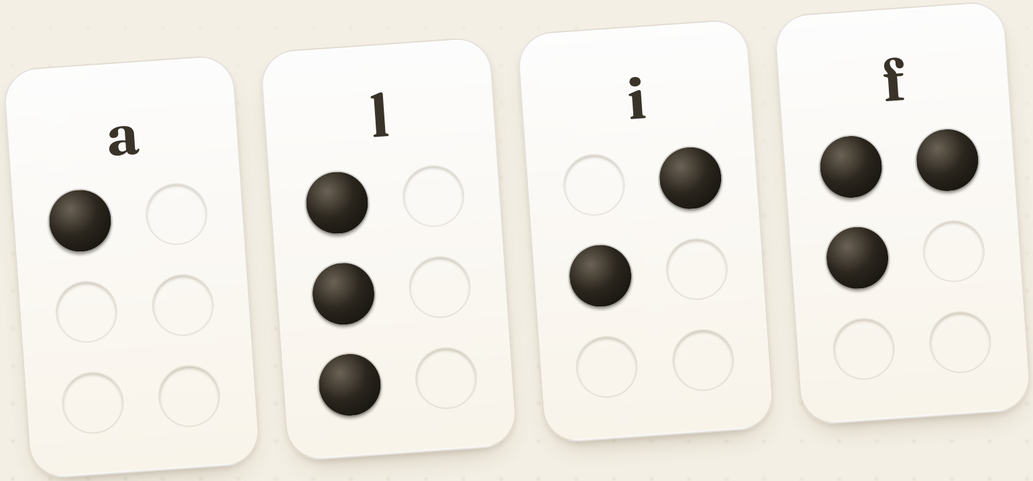
SXSW EDU · STUDENT IMPACT CHALLENGE 2027

Uzbek Latin in. *Braille* out.

AlifBraille is a free, open-source tool that gives blind students in Uzbekistan exam materials in the alphabet their exam is actually written in.

Elshod Kenjayev · Grade 11 · Presidential School in Tashkent, Uzbekistan



“alif” in 6-dot Braille: the first letter, the first step.

THE PROBLEM

The alphabet changed. Braille didn’t.

01

Exams moved to Latin

Uzbekistan’s schools, textbooks and university entrance exams now use the Uzbek Latin alphabet.

02

Braille stayed in Cyrillic

Most tactile books in specialized school libraries were produced in Cyrillic Braille.

03

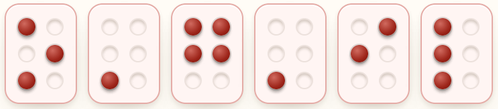
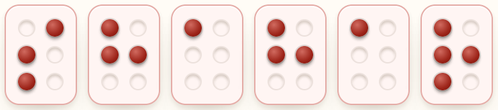
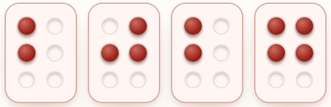
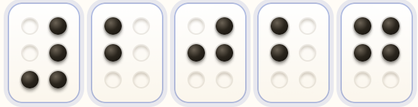
Teachers can’t check

Many teachers of blind students are sighted and don’t read Braille by touch, so they can’t verify a worksheet.

> *Blind students get fewer practice papers, later, in the wrong script, for the same exam everyone else takes.*

WHY EXISTING TOOLS FAIL

One wrong cell changes the answer.

UZBEK WORD	LETTER-BY-LETTER CONVERSION	ALIFBRAILLE	WHAT GOES WRONG
o‘g‘il “son”			Reads the apostrophes as punctuation. O‘ and G‘ are letters.
shahar “city”			Splits sh into two cells. In Uzbek it is one letter.
2027 “year”			No number sign, so it reads as the letters b-j-b-g.

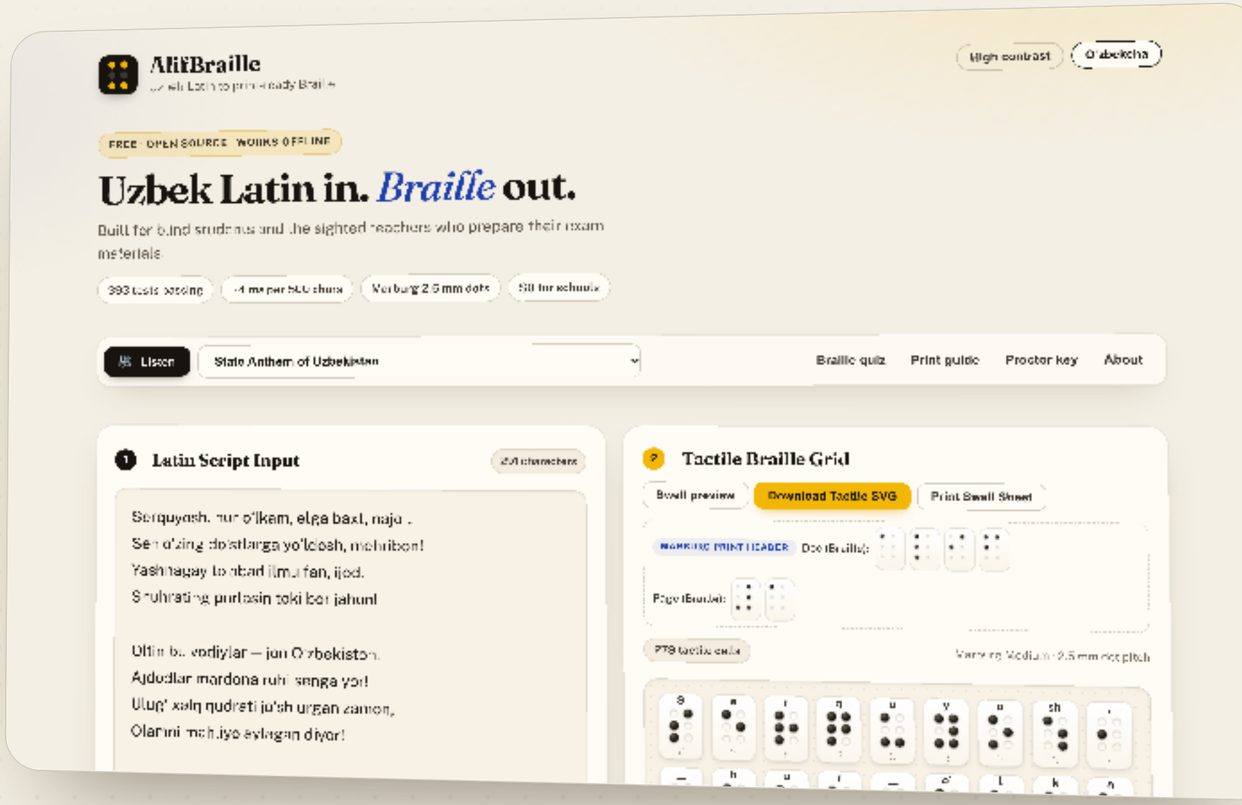
Braille shown is real output from the AlifBraille engine. Commercial Braille software doesn’t ship Uzbek Latin rules.

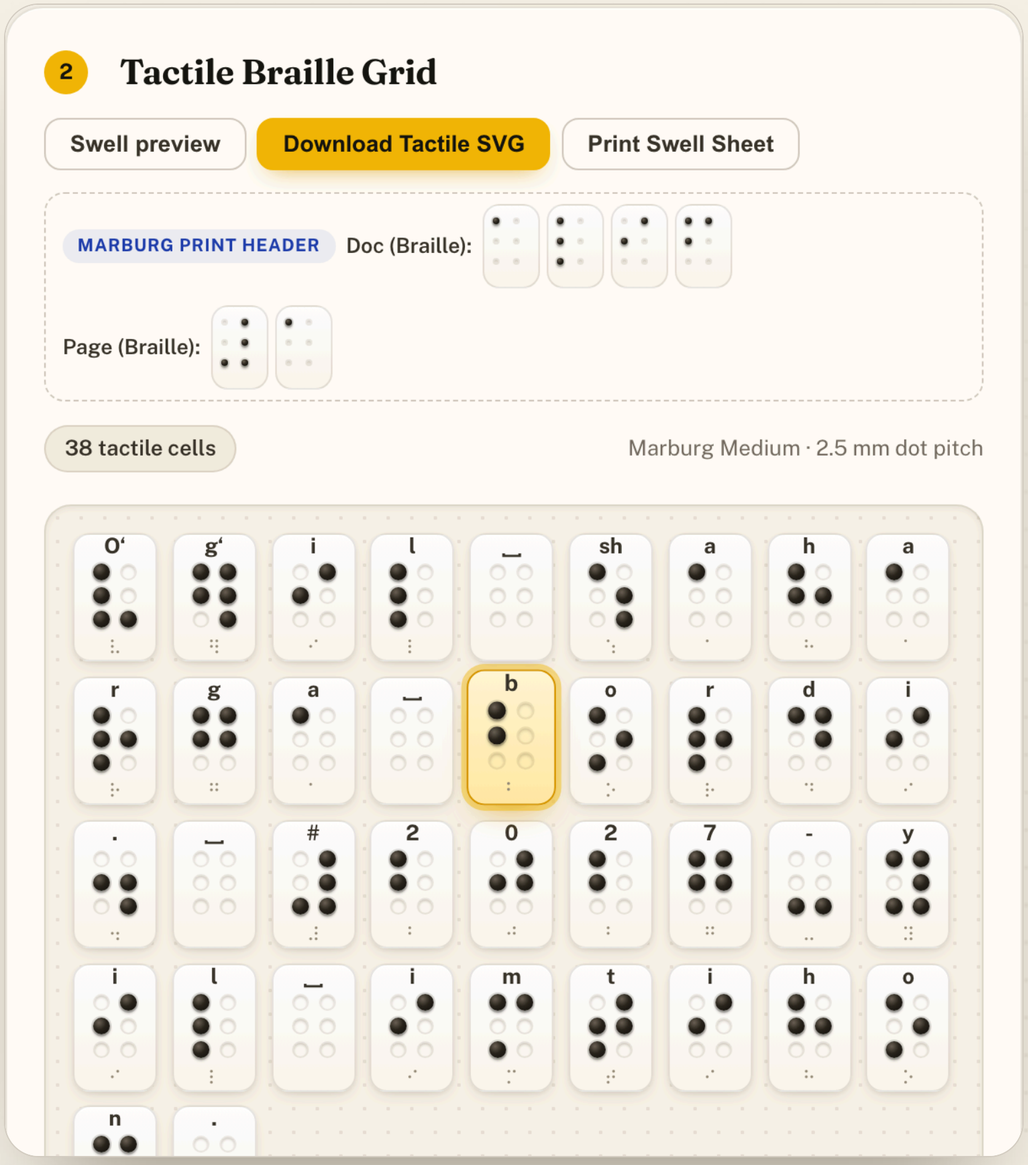
THE SOLUTION

Three steps. Any teacher. Any computer.

1. **Type or paste** Uzbek Latin text. Braille appears as you type, in about 4 ms.
2. **Check by ear.** The app reads aloud and lights up each cell as it speaks.
3. **Print** a calibrated A4 sheet for swell paper in one click.

Runs in the browser | Works offline | Free · MIT license





BUILT FOR SIGHTED TEACHERS

You don’t need to read Braille to check it.

As the text is read aloud, each Braille cell lights up in gold. A teacher hears “b” and sees the matching cell, so mistakes are caught on screen, not by a student’s fingertips on exam day.



sh = one cell
Uzbek letters like o‘, g‘, sh and ch each map to a single, correct cell.

PRINT-READY

Accurate to the millimetre.

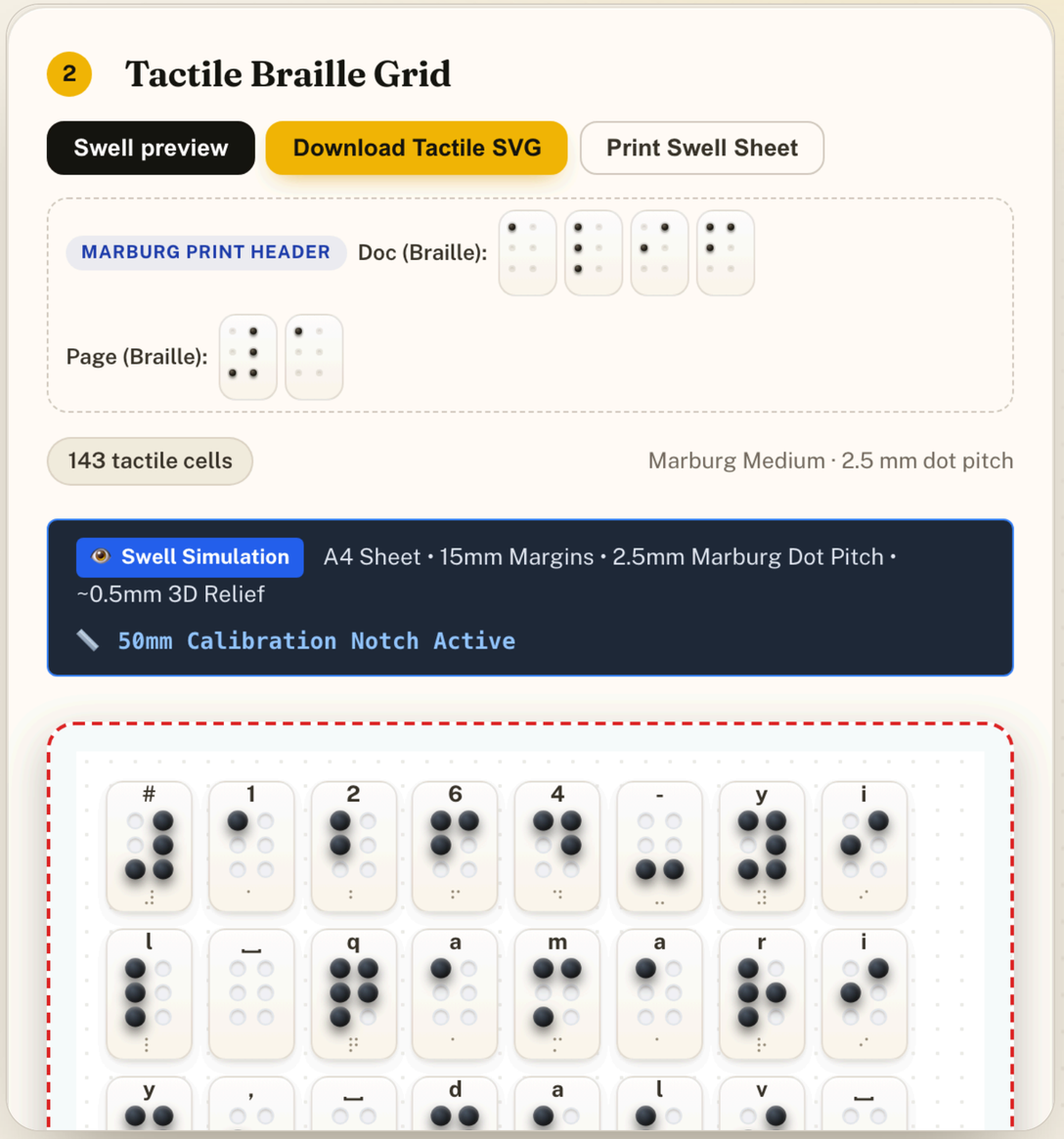
2.5 mm
dot pitch

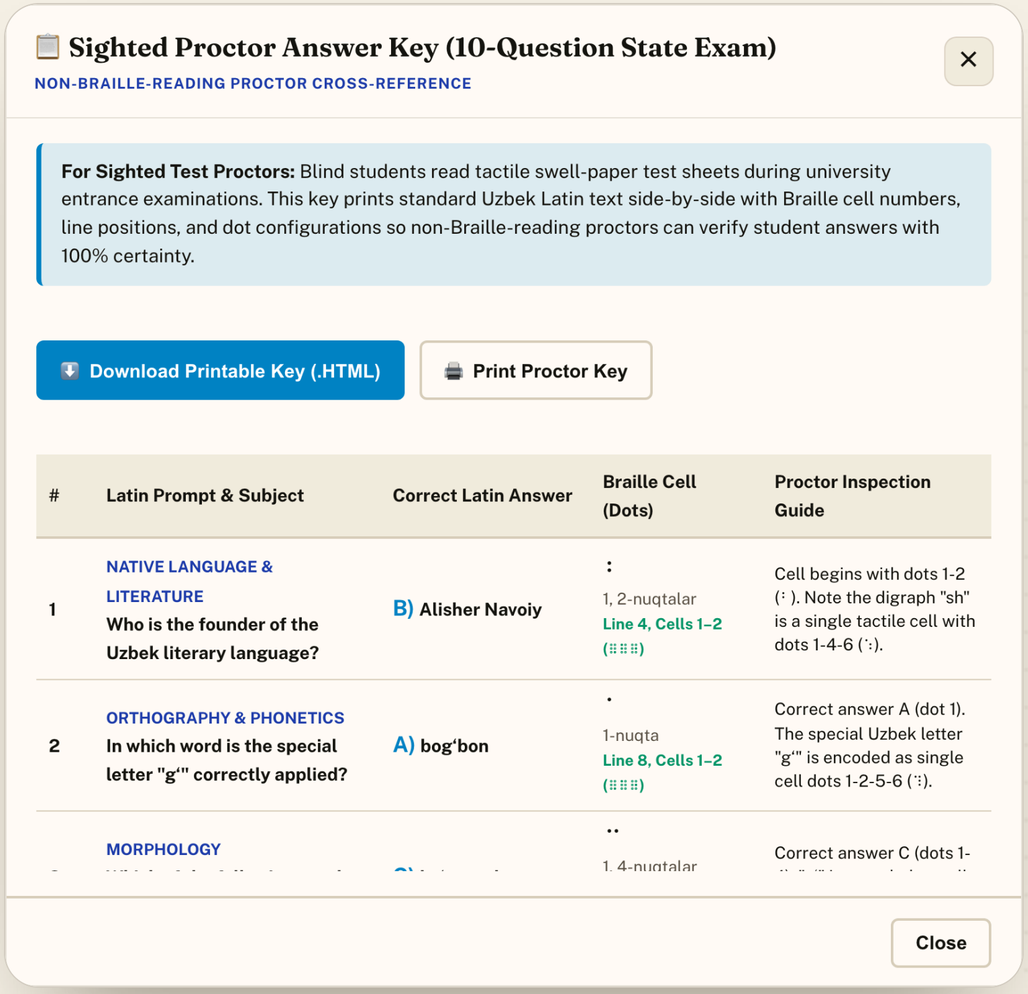
10 mm
line pitch

A4
15 mm margins

50 mm
ruler on every sheet

Sheets follow the Marburg Medium standard for swell paper. Printers often shrink pages without asking; the 50 mm ruler shows it **before** an expensive imported sheet goes through the heater.





READY FOR EXAM DAY

A full exam toolkit, not just a converter.

- **10-question state exam template** in the official multiple-choice format.
- **Proctor answer key** shows a sighted proctor which line and cell holds each answer.
- **Reverse Braille quiz** so teachers learn Braille alongside their students.
- **WCAG AAA high-contrast mode** for low-vision users.

WHERE IT STANDS TODAY

Working, tested, and honest about what's next.

393
automated tests passing

~4 ms
to convert 500 characters

<0.5 s
to load

AAA
WCAG contrast mode

100%
offline, in the browser

$0
for schools, MIT license

Done

- ✓ Live at braille.elshod.xyz
- ✓ Open source on GitHub
- ✓ Exam template, proctor key, quiz

Not yet

- ◌ Classroom pilot with blind students
- ◌ Review by Braille specialists
- ◌ Contracted (Grade 2) Braille

THE NEXT 12 MONTHS

From a working tool to every region.

MONTHS 1–3

Tashkent pilot

Convert exam practice sets with teachers at a school for visually impaired students.

MONTHS 4–6

Teachers & library

Workshops on convert → check → print, and a free library of practice sheets.

MONTHS 7–9

Regions

Samarkand, the Fergana Valley, Bukhara; more subjects.

MONTHS 10–12

New alphabets

Karakalpak and Kazakh Latin: the next countries changing scripts.

How we'll measure it: schools & teachers · pages converted · prep time per worksheet · errors caught per page · student reading accuracy

WHY ME · WHAT I'M ASKING FOR

Every blind student should practice in the same alphabet as their exam.

I'm preparing for the same Latin-script exams as Uzbekistan's blind students. I speak Uzbek, Russian and English, know both Uzbek alphabets, and built AlifBraille end to end on my own.

1. **A stage** to put this in front of educators who can adopt it.
2. **Introductions** to Braille specialists and schools for blind students.
3. **Mentorship** on scaling to other countries changing alphabets.

braille.elshod.xyz

github.com/Elshod09K/alifbraille

Elshod Kenjayev · Tashkent, Uzbekistan